MATH 3042- Quiz 3- Winter 2026- Solutions

MATH 3042- Quiz 3 Full Name:
Time: 20 minutes Set:

1) (4 marks) A server experiences **network errors** with probability 0.18 and **database errors** with probability 0.12. Assume the two types of errors are independent.

Find the probability of:

a) Both types of error occurring $P(N \cap D) = 0.18 \times 0.12 = 0.0216 = 2.16\%$

b) At least one type of error occurring

$P(N \cup D) = 0.18 + 0.12 - 0.0216 = 0.2784 = 27.84\%$

c) No error occurring

$P(\bar{N} \cap \bar{D}) = 1 - P(N \cup D) = 1 - 0.2784 = 0.7216 = 72.16\%$

2) (5 marks) An email filtering system has the following characteristics:

- 2% of incoming emails are spam
- The filter correctly blocks 95% of spam emails: $P(block|Spam) = 95\%$
- The filter incorrectly blocks 4% of legitimate emails: $P(block\ |Not\ Spam) = 4\%$

Answer the following:

1. If an email is blocked, what is the probability that it was spam?
2. What percentage of all emails are blocked by the filter?

Using this table is optional, alternatively you can use a formula or tree diagram

	Spam	**Legitimate**	
Blocked	1.9%	3.92%	5.82%
Not Blocked	0.1%	94.08%	94.18%
Sum:	**2%**	**98%**	

$P(block \cap spam) = 2\% \times 95\% = 1.9\%$
$P(not\ block \cap spam) = 2\% - 1.9\% = 0.1\%$

$P(block \cap not\ spam) = 4\% \times 98\% = 3.92\%$
$P(not\ block \cap not\ spam) = 98\% - 3.92\% = 94.08\%$

1. $P(spam|block) = \frac{1.9}{5.82} = 32.6\%$
2. $P(block) = 1.9 + 3.92 = 5.82\%$

3) (3 marks) A randomly generated API key consists of **10 characters**, chosen from:

- 26 lowercase letters
- 26 uppercase letters
- 10 digits

Calculate the probability that the key contains **at least one uppercase letter**.

All: 62^{10}
No uppercase: 36^{10}

$P(at\ least\ one\ upper\ case): 1 - \frac{36^{10}}{62^{10}} = 99.56\%$

4) (3 marks) Let Y be the number of security alerts raised per day, with:

$$P(Y = y) = \begin{cases} 0.40 & y = 0 \\ 0.30 & y = 1 \\ 0.20 & y = 2 \\ 0.10 & y = 3 \end{cases}$$

y	$P(y)$	$y.P(y)$	$y^2P(y)$
0	0.40	0	0
1	0.30	0.3	0.3
2	0.20	0.4	0.8
3	0.10	0.3	0.9
Sum	1	1	2

1. Compute $E[Y]$
2. Compute $\mathrm{Var}(Y)$

1. $E[Y] = 1\ \ alerts\ per\ day$
2. $Var[Y] = 2 - 1^2 = 1$